

Task Scheduling System using OS Concepts

Operating Systems Lab Mini Project

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1 Problem Statement

The objective of this project is to design and implement a task scheduling system that simulates core operating system functionalities such as concurrency, synchronization, inter-process communication, and role-based access control.

2 System Overview

The system follows a client-server architecture:

- Client
- Server
- Scheduler
- Worker

3 Implemented OS Concepts

3.1 Role-Based Authorization

Admin, User, and Guest roles are implemented with restricted access.

3.2 File Locking

File locking is implemented using `fcntl()`.

3.3 Concurrency Control

Threads are used for handling clients and scheduler.

3.4 Data Consistency

Ensured using locking and controlled updates.

3.5 Socket Programming

TCP client-server communication is implemented.

3.6 Inter-Process Communication (IPC)

Signals (SIGINT, SIGTERM) are used.

4 System Design

4.1 Task Format

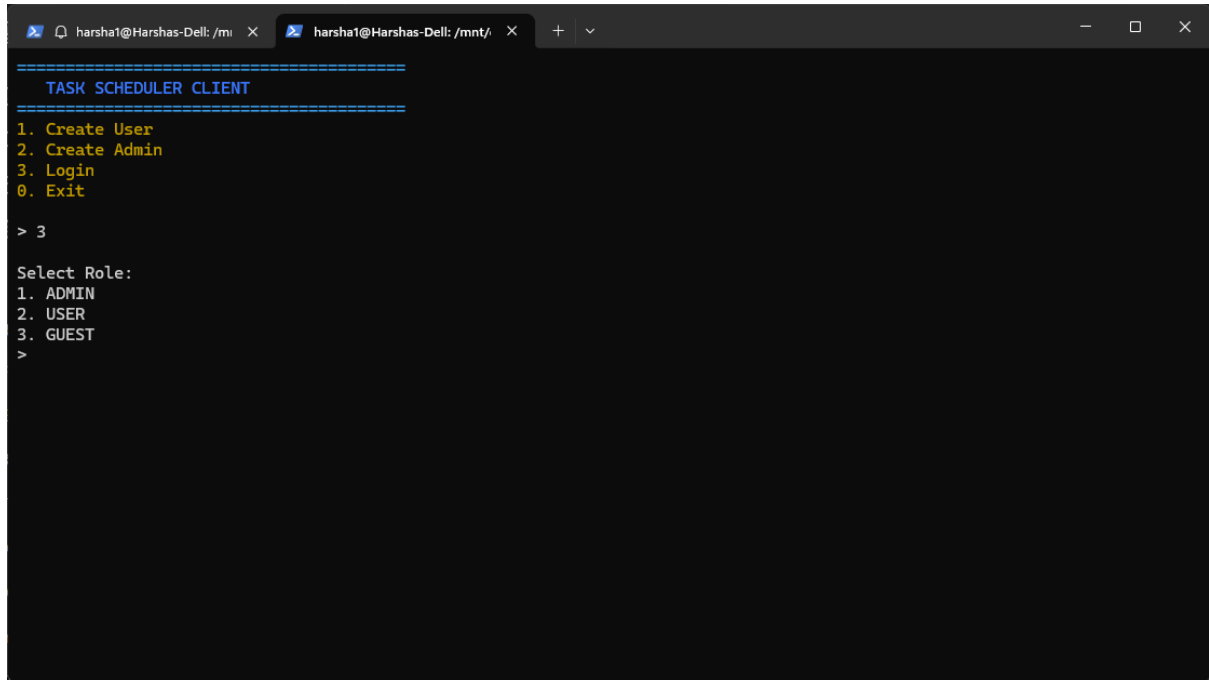
ID, filename, interval, created_at, last_run, priority, created_by

5 Key Features

- Multi-user system
- Scheduling with priority
- Logging
- Admin shutdown

6 Screenshots / Output

6.1 Client Login Interface



```
harsha1@Harshas-Dell: /mi  x  harsha1@Harshas-Dell: /mnt/  x  +  v  -  □  x
=====
TASK SCHEDULER CLIENT
=====
1. Create User
2. Create Admin
3. Login
0. Exit

> 3

Select Role:
1. ADMIN
2. USER
3. GUEST
>
```

Figure 1: Client interface showing role selection

6.2 Admin Dashboard

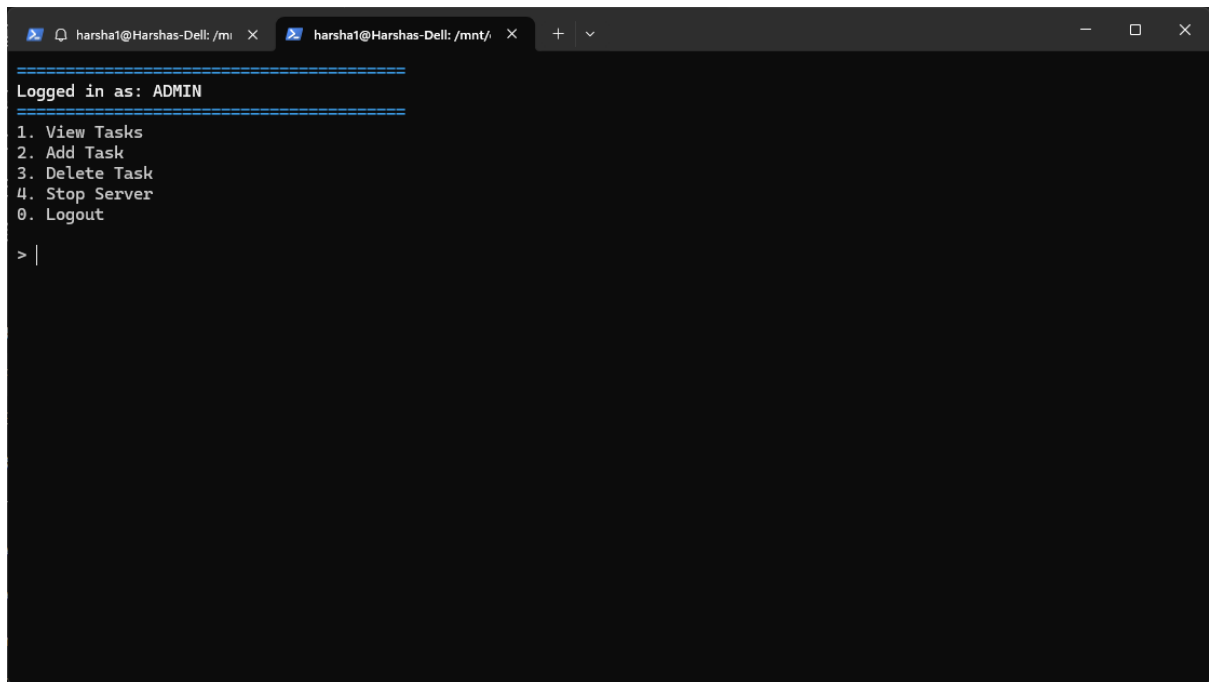


Figure 2: Admin menu with task operations and server shutdown

6.3 Task Viewing Output

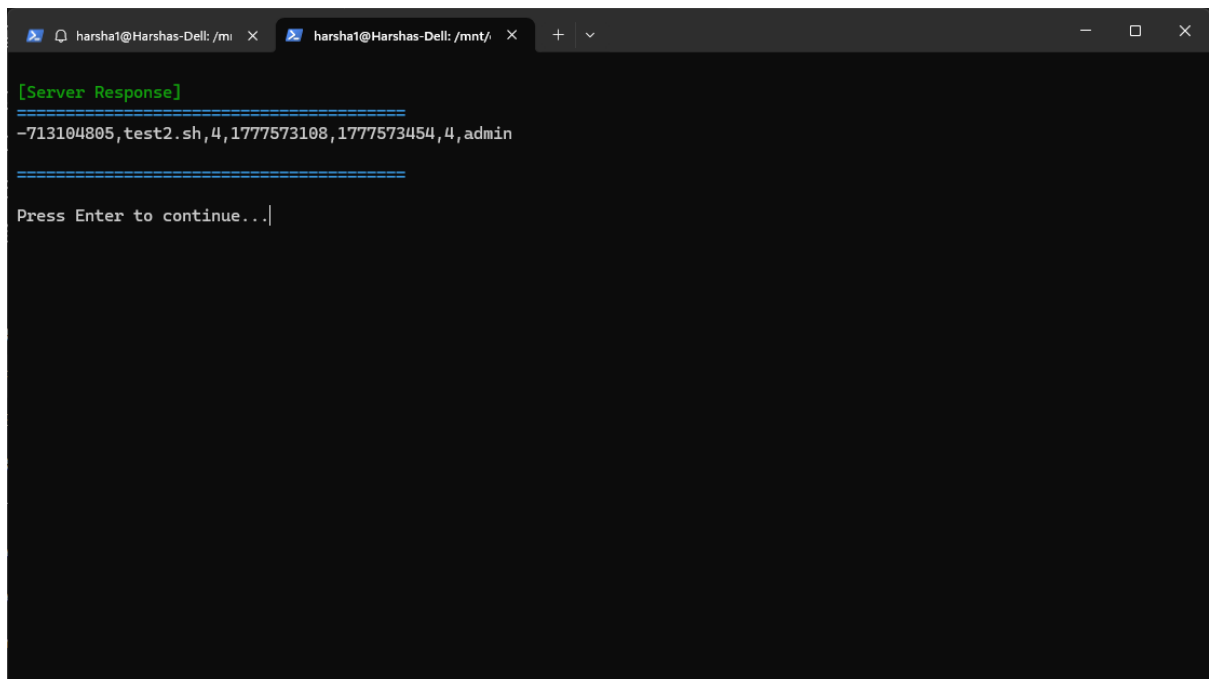


Figure 3: Viewing tasks stored in the system

7 Challenges Faced

- Handling concurrent file access
- Debugging socket communication
- Fixing task parsing bugs

8 Solutions

- File locking
- Structured parsing
- Logging system

9 Conclusion

This project demonstrates key OS concepts through a task scheduling system.

10 Future Work

- Database integration
- Distributed scheduling